CHAPTER 28

Quadratic Equations

INTRODUCTION

An equation of degree two is called a *quadratic equation*. The general form of a quadratic equation is $ax^2 + bx + c = 0$, where a, b, c are real numbers, $a \neq 0$ and x is a real variable. Some examples of quadratic equations are $x^2 + 4x + 3 = 0$, $3x^2 - 4x + 5 = 0$ and $3x^2 + 2x - 3 = 0$.

Roots of a Quadratic Equation

A *root* of the equation $f(x) = 0$ is that value of x which makes $f(x) = 0$. In other words, $x = a$ is said to be a root of $f(x) = 0$, where $f(a)$ is the value of the polynomial $f(x)$ at $x = a$ and is obtained by replacing x by a in $f(x)$.

For example, -1 is a root of the quadratic equation $x^2 + 6x + 5 = 0$ because $(-1)^2 + 6(-1) + 5 = 0$.

Solution of a Quadratic Equation

If there is a quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$, the roots of this equation are

$$\frac{-b+\sqrt{b^2-4ac}}{2a} \text{ and } \frac{-b-\sqrt{b^2-4ac}}{2a}.$$

Illustration 1: Solve the following quadratic equations:

(i) $6x^2 + x - 2 = 0$

(ii) $2x^2 + x - 1 = 0$

Solution: (i) Using formula:

$$\text{The roots are } x = \frac{-1\pm\sqrt{(1)^2-4(6)(-2)}}{2\times 6}$$

$$= \frac{-1\pm\sqrt{49}}{12} = \frac{6}{12}, \frac{-8}{12}.$$

i.e., $\frac{1}{2}, \frac{-2}{3}$.

Using factorization:

$$6x^2 + x - 2 = 0 \Leftrightarrow 6x^2 + 4x - 3x - 2 = 0$$

$$\Leftrightarrow 2x(3x + 2) - 1(3x + 2) = 0$$

$$\Leftrightarrow (2x + 1)(3x + 2) = 0$$

$$\Leftrightarrow x = \frac{1}{2} \text{ or } x = -\frac{2}{3}.$$

(ii) Using formula:

$$\text{The roots are } x = \frac{-1\pm\sqrt{(1)^2-4(2)(-1)}}{2\times 2}$$

$$= \frac{-1\pm\sqrt{9}}{4} = \frac{-1\pm 3}{4}$$

$$= \frac{2}{4}, \frac{-4}{4} \text{ i.e., } \frac{1}{2}, -1.$$

Using factorization:

$$2x^2 + x - 1 = 0 \Leftrightarrow 2x^2 + 2x - x - 1 = 0$$

$$\Leftrightarrow 2x(x + 1) - 1(x + 1) = 0$$

$$\Leftrightarrow (2x - 1)(x - 1) = 0.$$

$$\Leftrightarrow x = \frac{1}{2} \text{ or } x = -1.$$

Nature of Roots

A quadratic equation has exactly two roots may be real or imaginary or coincident.

If $ax^2 + bx + c$, $a \neq 0$, then $D = b^2 - 4ac$ is called *discriminant*.

1. If $D > 0$, then there are two distinct and real roots given by

$$\alpha = \frac{-b+\sqrt{b^2-4ac}}{2a}, \ \beta = \frac{-b-\sqrt{b^2-4ac}}{2a}.$$

2. If $D = 0$, then there is a repeated real root given by

$$\alpha = -\frac{b}{2a} \text{ i.e., roots are real and equal.}$$

3. If $D < 0$, then there are no real roots.

Notes

The roots are rational if $D > 0$ and D is a perfect square whereas the roots are irrational if $D > 0$ but D is not a perfect square.

Illustration 2: Find the nature of the roots of the equations:

(i) $2x^2 + x - 1 = 0$

(ii) $x^2 + x + 1 = 0$

(iii) $x^2 + 5x + 5 = 0$

(iv) $\frac{4}{3}x^2 - 2x + \frac{3}{4} = 0$

Solution: (i) $D = (1)^2 - 4 \times 2 \times (-1) = 9 > 0$.

Also, D is a perfect square.

So, the roots are real, distinct and rational.

(ii) $D = (1)^2 - 4 \times 1 \times 1 = -3 < 0$

So, the roots are imaginary.

(iii) $D = (5)^2 - 4 \times 1 \times 5 = 5 > 0$.

Also, D is not a perfect square.

So, the roots are real, distinct and irrational.

(iv) $D = (-2)^2 - 4 \times \frac{4}{3} \times \frac{3}{4} = 0$.

So, the roots are real and equal.

Illustration 3: For what value of k will the quadratic equation $kx^2 - 2\sqrt{5}\,x + 4 = 0$ have real and equal roots.

Solution: $D = (-2\sqrt{5})^2 - 4 \times k \times 4 = 20 - 16k$.

The given equation will have real and equal roots if $D = 0$.

i.e., $20 - 16k = 0$ or $k = \frac{20}{16} = \frac{5}{4}$.

Notes

1. If $p + \sqrt{q}$ is a root of a quadratic equation, then its other root is $p - \sqrt{q}$.

Illustration 4: If $2 + \sqrt{3}$ is one root of a quadratic equation, find the other root.

Solution: The other root is $2 - \sqrt{3}$.

2. $ax^2 + bx + c$ can be expressed as a product of two linear factors only when $D \geq 0$.

Illustration 5: For what value of k, the quadratic polynomial $kx^2 + 4x + 1$ can be factorized into two real linear factors.

Solution: $D = (4)^2 - 4 \times k \times 1 = 16 - 4k$.

The given quadratic polynomial can be factorized into real linear factors if $D \geq 0$.

i.e., $16 - 4k \geq 0$ or $-4k \geq -6$ or $k \leq 4$.

Relation Between Roots and Coefficients

Let, α, β be the roots of the equation,

$$ax^2 + bx + c = 0$$

Then, sum of the roots

$$= \alpha + \beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$$

and product of the roots

$$= \alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

Illustration 6: Find the sum and the product of the roots of the quadratic equation $2x^2 + 5\sqrt{3}\,x + 6 = 0$.

Solution: Here $a = 2$, $b = 5\sqrt{3}$, $c = 6$.

$$\therefore \quad \text{Sum of the roots} = -\frac{b}{a} = -\frac{5\sqrt{3}}{2}.$$

$$\text{Product of the roots} = \frac{c}{a} = \frac{6}{2} = 3.$$

Formation of a Quadratic Equation with Given Roots

If α, β are the roots of a quadratic equation the equation can he written as

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

i.e., $x^2 - (\text{sum of roots})x + \text{product of roots} = 0$.

Illustration 7: Find the quadratic equation whose roots are 5 and -6.

Solution: Sum of roots $= 5 + (-6) = -1$,

Product of roots $= 5 \times (-6) = -30$.

$\therefore$ The required quadratic equation is

$x^2 - (-1)x + (-30) = 0$ i.e., $x^2 + x - 30 = 0$.

EXERCISE–I

1. In the following determine the set of value of P for which the given quadratic equation has real roots.

$$Px^2 + 4x + 1 = 0$$

(a) $P \neq 4$ (b) $P > 4$
(c) $P \leq 4$ (d) $P \geq 4$

2. If one root of the quadratic equation $2x^2 + Px + 4 = 0$ is 2, find the second root and value of P.

(a) 1, −6 (b) 1, 6
(c) −1, 6 (d) −1, −6

3. One root of the quadratic equation $x^2 - 5x + 6 = 0$ is 3. Find the other root.

(a) 2 (b) −2
(c) 1 (d) −1

4. The roots of the equation

$\sqrt{7}x^2 - 6x - 13\sqrt{7} = 0$ are

(a) $-\sqrt{7}, \frac{-13\sqrt{7}}{7}$ (b) $\sqrt{7}, \frac{-13\sqrt{7}}{7}$

(c) $-\sqrt{7}, \frac{13\sqrt{7}}{7}$ (d) None of these

5. The roots of the equation

$3a^2x^2 - abx - 2b^2 = 0$ are

(a) $\frac{b}{a}, \frac{-2b}{3a}$ (b) $\frac{b}{a}, \frac{2b}{3a}$

(c) $\frac{-b}{a}, \frac{-2b}{3a}$ (d) None of these

6. The roots of the equation

$a^2x^2 - 3abx - 2b^2 = 0$ are

(a) $\frac{2b}{a}, \frac{-b}{a}$ (b) $\frac{2b}{a}, \frac{b}{a}$

(c) $\frac{-2b}{a}, \frac{b}{a}$ (d) None of these

7. Construct a quadratic equation whose roots are $\sqrt{2}$ and $2\sqrt{2}$

(a) $x^2 - 3\sqrt{2}x - 4 = 0$ (b) $x^2 - 3\sqrt{2}x + 4 = 0$

(c) $x^2 + 3\sqrt{2}x - 4 = 0$ (d) $x^2 + 3\sqrt{2}x + 4 = 0$

8. The roots of the equation

$ax^2 + (4a^2 - 3b)x - 12ab = 0$ are

(a) $4a, \frac{3b}{a}$ (b) $-4a, \frac{3b}{a}$

(c) $-4a, \frac{3b}{a}$ (d) $-4a, \frac{-3b}{a}$

9. Construct a quadratic equation whose roots have the sum = 6 and product = −16.

(a) $x^2 - 6x - 16 = 0$ (b) $x^2 + 6x - 16 = 0$

(c) $x^2 - \sqrt{3}x - 6 = 0$ (d) None of these

10. In the following, find the value (s) of P so that the given equation has equal roots

$3x^2 - 5x + P = 0$

(a) −25/12 (b) 25/6
(c) 25/12 (d) − 25/6

11. If α and β are the roots of the equation $ax^2 + bx + c = 0$, find the value of $\alpha^2 + \beta^2$

(a) $\frac{b^2 - 2ac}{2a^2}$ (b) $\frac{b^2 + 2ac}{a^2}$

(c) $\frac{b^2 + 2ac}{2a^2}$ (d) $\frac{b^2 - 2ac}{a^2}$

12. If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, the value of $\alpha^3 + \beta^3$ is

(a) $\frac{b(b^2 - 3ac)}{a^3}$ (b) $\frac{b(3ac - b^2)}{a^3}$

(c) $\frac{b(3ac + b^2)}{a^3}$ (d) None of these

13. If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is

(a) $\frac{b^2 - 2ac}{ac}$ (b) $\frac{b^2 - 2ac}{2ac}$

(c) $\frac{b^2 - ac}{2ac}$ (d) $\frac{b^2 + 2ac}{ac}$

14. The quadratic equation with rational coefficients, whose one root is $\sqrt{5}$, is:

(a) $x^2 + 5 = 0$ (b) $x^2 - 10 = 0$
(c) $x^2 - 5 = 0$ (d) None of these

15. The equation $x^2 - px + q = 0$, $p, q \in R$ has on real root if:

(a) $p^2 \le 4q$ (b) $p^2 < 4q$
(c) $p^2 > 4q$ (d) None of these

16. Determine p so that the equation $x^2 + 5px + 16 = 0$ has on real root.

(a) $\frac{-4}{5} < p < \frac{4}{5}$ (b) $\frac{-8}{5} < p < \frac{8}{5}$

(c) $p < -\frac{4}{5}$ or $p > \frac{4}{5}$ (d) None of these

17. For what value of k the quadratic polynomial $3z^2 + 5z + k$ can be factored into product of real linear factors?

(a) $k \le \frac{25}{6}$ (b) $k \le \frac{25}{12}$

(c) $k \ge \frac{25}{12}$ (d) $k \ge \frac{25}{6}$

18. $x = 3$ i a solution of the equation $3x^2 + (k - 1)x + 9 = 0$ if k has value

(a) 13 (b) –13
(c) 11 (d) –11

19. One root of the equation $3x^2 - 10x + 3 = 0$ is $\frac{1}{3}$. Find the other root.

(a) 3 (b) 1/3
(c) –3 (d) None of these

20. The expression $x^4 + 7x^2 + 16$ can be factored as:

(a) $(x^2 + x + 1)(x^2 + x + 16)$

(b) $(x^2 + x + 1)(x^2 - x + 16)$

(c) $(x^2 + x + 4)(x^2 - x + 4)$

(d) $(x^2 + x - 4)(x^2 - x - 4)$

21. The common root of the equations $x^2 - 7x + 10 = 0$ and $x^2 - 10x + 16 = 0$ is:

(a) –2 (b) 3
(c) 5 (d) 2

22. The roots of the equation $x^2 + px + q = 0$ are equal if:

(a) $p^2 = 2q$ (b) $p^2 = 4q$
(c) $p^2 = -4q$ (d) $p^2 = -2q$

23. An equation equivalent to the quadratic equation $x^2 - 6x + 5 = 0$ is:

(a) $6x^2 - 5x + 1 = 0$ (b) $x^2 - 5x + 6 = 0$
(c) $5x^2 - 6x + 1 = 0$ (d) $|x - 3| = 2$

24. Divide 16 into 2 parts such the twice the square of the larger part exceeds the square of the smaller part by 164

(a) 10, 6 (b) 8, 8
(c) 12, 4 (d) None of these

25. With respect to the roots of $x^2 - x - 2 - 0$, we can say that:

(a) both of them are natural numbers
(b) both of them are integers
(c) the latter of the two is negative
(d) None of these

26. The solution of $2 - x = \frac{x - 2}{x}$ would include:

(a) –2, –1 (b) 2, –1
(c) –4, 2 (d) 4, –2

27. If $\log_{10}(x^2 - 6x + 45) = 2$, then the values of x are

(a) 6, 9 (b) 9, –5
(c) 10, 5 (d) 11, –5

28. If α, β are the roots of the equation $x^2 - 5x + 6 = 0$, construct a quadratic equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.

(a) $6x^2 + 5x - 1 = 0$ (b) $6x^2 - 5x - 1 = 0$
(c) $6x^2 - 5x + 1 = 0$ (d) $6x^2 + 5x + 1 = 0$

29. The roots of $\frac{x+4}{x-4} + \frac{x-4}{x+4} = \frac{10}{3}$ are:

(a) ± 4 (b) ± 6
(c) ± 8 (d) $2 \pm \sqrt{3}$

30. The roots of the equation $ax^2 + bx + c = 0$ will be reciprocal if

(a) $a = b$ (b) $b = c$
(c) $c = a$ (d) None of these

31. Form a quadratic equation whose one root is $3-\sqrt{5}$ and the sum of roots is 6

(a) $x^2 - 6x + 4 = 0$ (b) $x^2 + 6x + 4 = 0$
(c) $x^2 - 6x - 4 = 0$ (d) None of these

32. The value of k for which the roots α, β of the equation: $x^2 - 6x + k = 0$ satisfy the relation $3\alpha + 2\beta = 20$, is

(a) 8 (b) −8
(c) 16 (d) −16

33. Find two consecutive positive odd integers whose squares have the sum 290.

(a) 11, 13 (b) 13, 15
(c) 9, 11 (d) None of these

34. Consider the equation $px^2 + qx + r = 0$, where p, q, r are real. The roots are equal in magnitude but opposite in sign when:

(a) $q = 0$, $r = 0$, $p \neq 0$
(b) $p = 0$, $qr \neq 0$
(c) $r = 0$, $pr \neq 0$
(d) $q = 0$, $pr \neq 0$

35. Determine k such that the quadratic equation $x^2 - 2(1+3k)x + 7(3+2k) = 0$ has equal roots.

(a) $2, \frac{-10}{9}$ (b) $2, \frac{10}{9}$

(c) $-2, \frac{10}{9}$ (d) $-2, \frac{-10}{9}$

36. If the equations $x^2 + 2x - 3 = 0$ and $x^2 + 3x - k = 0$ have a common root, then the non–zero value of k is:

(a) 1 (b) 2
(c) 3 (d) 4

37. The roots of the equation $4x - 3.2x^2 + 32 = 0$ would include:

(a) 1, 2 and 3 (b) 1 and 2
(c) 1 and 3 (d) 2 and 3

38. The positive value of m for which the roots of the equation $12x^2 + mx + 5 = 0$ are in the ratio 3:2 is:

(a) $5\sqrt{10}$ (b) $\frac{5}{2}\sqrt{10}$

(c) $\frac{5}{12}$ (d) $\frac{12}{5}$

39. If α, β are the roots of the equation $2x^2 - 3x + 1 = 0$, form an equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

(a) $2x^2 + 5x + 2 = 0$ (b) $2x^2 - 5x - 2 = 0$
(c) $2x^2 - 5x + 2 = 0$ (d) None of these

40. Find the quadratic equation whose roots are reciprocal of the roots of the equation

$3x^2 - 20x + 17 = 0$

(a) $17x^2 - 20x + 3 = 0$
(b) $17x^2 + 20x + 3 = 0$
(c) $17x^2 - 20x - 3 = 0$
(d) None of these

41. If α and β are the roots of the equation

$x^2 - 3\lambda x + \lambda^2 = 0$, find λ if $\alpha^2 + \beta^2 = \frac{7}{4}$

(a) $\pm\frac{1}{2}$ (b) $\pm\frac{\sqrt{7}}{2}$

(c) $\pm\frac{\sqrt{3}}{2}$ (d) None of these

42. If α, β are the roots of the equation $ax^2 + bx + b = 0$, then

$\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} =$

(a) 1 (b) 0
(c) 2 (d) 3

43. The expression $x^2 - x + 1$ has:

(a) one proper linear factor
(b) two proper linear factors
(c) no proper linear factor
(d) None of these

44. The length of a rectangular plot is 8 m greater than its breadth. If the area of the plot is 308 m^2, find the length of the plot.

(a) 22 m (b) 18 m
(c) 20 m (d) None of these

45. If α, β are the roots of the equation $x^2 + kx + 12 = 0$ such that $\alpha - \beta = 1$, the value of k is

(a) 0 (b) ± 5
(c) ± 1 (d) ± 7

46. The value of x in the equation

$$\left(x+\frac{1}{x}\right)^2-\frac{3}{2}\left(x-\frac{1}{x}\right)=4 \text{ is:}$$

(a) -2 (b) $\frac{1}{2}$

(c) -1 (d) 0

47. If α, β are the roots of the quadratic equation $x^2 - 8x + k = 0$, find the value of k such the $\alpha^2 + \beta^2 = 40$

(a) 12 (b) 14

(c) 10 (d) 16

48. Find the value of k so that the sum of the roots of the equation $3x^2 + (2x + 1)x - k - 5 = 0$ is equal to the product of the roots:

(a) 4 (b) 6

(c) 2 (d) 8

EXERCISE–2
(BASED ON MEMORY)

1. Which of the following equations has equal roots?

(a) $3x^2 - 6x + 2 = 0$
(b) $3x^2 - 6x + 3 = 0$
(c) $x^2 - 8x + 8 = 0$
(d) $8x^2 - 8x + 2 = 0$

[SSC CHSL (10+2) Tier-I (CBE), 2017]

2. If sum of the roots of a quadratic equation is 1 and product of the roots is –20. Find the quadratic equation.

(a) $x^2 - x - 20 = 0$
(b) $x + x + 20 = 0$
(c) $x^2 + x - 20 = 0$
(d) $x^2 - x + 20 = 0$

[SSC CGL Tier-I CBE, 2017]

3. If $a(x + y) - b(x - y) = 2ab$ then the value of $2(x^2 + y^2)$ is:

(a) $2(a^2-b^2)$ (b) $2(a^2+b^2)$

(c) $4(a^2-b^2)$ (d) $4(a^2+b^2)$

[SSC CGL Tier-I (CBE), 2016]

4. If $(x - 2)(x - p) = x^2 - ax + 6$, then the value of $(a - p)$ is

(a) 0 (b) 1

(c) 2 (d) 3

[SSC CGL Tier-I (CBE), 2016]

5. If $a^2 = by + cz$, $b^2 = cz + zx$, $c^2 = ax + by$, then the value of $\frac{x}{a+x}+\frac{y}{b+y}+\frac{z}{c+z}$ is

(a) 1 (b) $a + b + c$

(c) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ (d) 0

[SSC CGL Tier-I (CBE), 2016]

6. If $x = 332$, $y = 333$, $z = 335$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is

(a) 7000 (b) 8000

(c) 9000 (d) 10000

[SSC, 2015]

7. If $m = -4$, $n = -2$, then the value of $m^3 - 3m^2 + 3m + 3n + 3n^2 + n^3$ is

(a) 124 (b) −124

(c) 126 (d) −126

[SSC, 2015]

8. If $\frac{m-a^2}{b^2+c^2}+\frac{m-b^2}{c^2+a^2}+\frac{m-c^2}{a^2+b^2}=3$, then the value of m is

(a) $a^2 + b^2$ (b) $a^2 + b^2 + c^2$

(c) $a^2 - b^2 - c^2$ (d) $a^2 + b^2 - c^2$

[SSC, 2015]

9. Let $x=\frac{\sqrt{13}+\sqrt{11}}{\sqrt{13}-\sqrt{11}}$ and $y=\frac{1}{x}$, then the value of $3x^2-5xy+3y^2$ is

(a) 1717 (b) 1171

(c) 1771 (d) 1177

[SSC, 2015]

10. If $x^2+y^2+z^2=xy+yz+zx$, then the value of $\frac{3x^4+7y^4+5z^4}{5x^2y^2+7y^2z^2+3z^2x^2}$ is

(a) 0 (b) −1

(c) 2 (d) 1

[SSC, 2015]

11. If $x-\sqrt{3}-\sqrt{2}=0$ and $y-\sqrt{3}+\sqrt{2}=0$, then value of $\left(x^3-20\sqrt{2}\right)-\left(y^3+2\sqrt{2}\right)$

(a) 1 (b) 0
(c) 2 (d) 3

[SSC, 2015]

12. If $3(a^2 + b^2 + c^2) = (a + b + c)^2$, then the relation between a, b and c is

(a) $a \neq b \neq c$ (b) $a = b \neq c$
(c) $a = b = c$ (d) $a \neq b = c$

[SSC, 2015]

13. If $(x^3-y^3):(x^2+xy+y^2)=5:1$ and $(x^2-y^2):(x-y)$ = 7 : 1, then the ratio $2x:3y$ equals

(a) 4 : 3 (b) 4 : 1
(c) 2 : 3 (d) 3 : 2

[SSC, 2015]

14. If $x=a^{\frac{1}{2}}+a^{-\frac{1}{2}}, y=a^{\frac{1}{2}}-a^{-\frac{1}{2}}$, then the value of $(x^4-x^2y^2-1)+(y^4-x^2y^2+1)$

(a) 12 (b) 14
(c) 16 (d) 13

[SSC, 2015]

15. If $a + b = 1$, find the value of $a^3 + b^3 - ab - (a^2 - b^2)^2$

(a) 2 (b) 1
(c) 0 (d) –1

[SSC, 2015]

16. If $a-\frac{1}{a-3}=5$, then the value of $(a-3)^3-\frac{1}{(a-3)^3}$ is

(a) 7 (b) 2
(c) 5 (d) 14

[SSC, 2015]

17. If $\left(\frac{p^{-1}q^2}{p^3q^{-2}}\right)^{\frac{1}{3}}\div\left(\frac{p^5q^{-3}}{p^{-2}q^3}\right)^{\frac{1}{3}}=p^aq^b$, then the value of $a + b$, where p and q are different positive primes, is

(a) 1 (b) –1
(c) 2 (d) 0

[SSC, 2015]

18. If $(3x-2y):(2x+3y)=5:6$, then one of the value of $\left(\frac{\sqrt[3]{x}+\sqrt[3]{y}}{\sqrt[3]{x}-\sqrt[3]{y}}\right)^2$ is

(a) 5 (b) 25
(c) $\frac{1}{5}$ (d) $\frac{1}{25}$

[SSC, 2015]

19. If m – 5n, = 2, then the value of (m^3 – $125n^3$ – 30mn) is

(a) 7 (b) 8
(c) 9 (d) 6

20. Given that $x^3 + y^3 = 72$ and $xy = 6$ with $x > y$, then the value of $x - y$ is:

(a) 2 (b) –2
(c) –4 (d) 4

[SSC, 2015]

21. If $x+\frac{1}{x}=2$, then the value of $x^{12}-\frac{1}{x^{12}}$ is:

(a) 4 (b) 2
(c) 0 (d) – 4

[SSC, 2015]

22. If $x=\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ and $y=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$ then the value of: $\frac{x^2+xy+y^2}{x^2-xy+y^2}$ = ?

(a) $\frac{65}{63}$ (b) $\frac{67}{65}$
(c) $\frac{63}{61}$ (d) $\frac{69}{67}$

[SSC, 2015]

23. If $x^2 + x = 5$ then the value of: $(x+3)^3+\frac{1}{(x+3)^3}$

(a) 120 (b) 130
(c) 140 (d) 110

[SSC, 2015]

24. If $4a-\frac{4}{a}+3=0$ then the value of: $a^3-\frac{1}{a^3}+3=?$

(a) $\frac{7}{16}$ (b) $\frac{21}{64}$

(c) $\frac{3}{16}$ (d) $\frac{21}{16}$

[SSC, 2015]

25. If $x = z = 225$ and $y = 226$ then the value of: $x^3 + y^3 + z^3 - 3xyz$

(a) 576 (b) 674
(c) 765 (d) 676

[SSC, 2015]

26. If $x^2 + y^2 + z^2 = 2(x + z - 1)$. Then the value of: $x^3 + y^3 + z^3 = ?$

(a) 1 (b) –1
(c) 0 (d) 2

[SSC, 2015]

27. If $5x + 9y = 5$ and $125x^3 + 729y^3 = 120$ then the value of the product of x and y is

(a) 135 (b) $\frac{1}{135}$
(c) $\frac{1}{9}$ (d) 45

[SSC, 2015]

28. The factors of $(a^2 + 4b^2 + 4b - 4ab - 2a - 8)$ are:

(a) $(a-2b-4)(a-2b+2)$

(b) $(a-b+2)(a+4b+4)$

(c) $(a+2b-4)(a+2b+2)$

(d) $(a+2b-1)(a-2b+1)$

[SSC, 2014]

Directions (29–33): In this question two equations numbered I and II are given. You have to solve both the equations and find out the correct option.

29. I. $6x^2 + 41x + 63 = 0$

II. $4y^2 + 8y + 3 = 0$

(a) Relationship between x and y cannot be established
(b) $x \geq y$
(c) $x < y$
(d) $x > y$
(e) $x \leq y$

[IBPS PO/MT, 2014]

30. I. $x^2 + 10x + 24 = 0$

II. $4y^2 - 17y + 18 = 0$

(a) $x \leq y$
(b) $x \geq y$
(c) Relationship between x and y cannot be established
(d) $x > y$
(e) $x < y$

[IBPS PO/MT, 2014]

31. I. $24x^2 + 38x + 15 = 0$

II. $12y^2 + 28y + 15 = 0$

(a) $x \leq y$ (b) $x > y$
(c) $x \geq y$ (d) $x < y$
(e) $x = y$, or Relationship between x and y cannot be established

[IBPS PO/MT, 2014]

32. I. $3x^2 - 20x - 32 = 0$

II. $2y^2 - 3y - 20 = 0$

(a) $x < y$
(b) $x \leq y$
(c) $x > y$
(d) Relationship between x and y cannot be established
(e) $x \geq y$

[IBPS PO/MT, 2014]

33. I. $x^2 - 20x + 91 = 0$

II. $y^2 - 32y + 247 = 0$

(a) $x > y$
(b) Relationship between x and y cannot be established
(c) $x \geq y$
(d) $x \leq y$
(e) $x < y$

[IBPS PO/MT, 2014]

34. If $x + \frac{1}{x} = 5$, then $x^6 + \frac{1}{x^6}$ is

(a) 12098 (b) 12048
(c) 14062 (d) 12092

[SSC, 2014]

35. If $x^2 - 3x + 1 = 0$, then the value of $\frac{x^6 + x^4 + x^2 + 1}{x^3}$ will be

(a) 18 (b) 15
(c) 21 (d) 30

[SSC, 2014]

36. If $x^4 + \frac{1}{x^4} = 119$ & $x > 1$, then find the positive value of $x^3 - \frac{1}{x^3}$

(a) 25 (b) 27
(c) 36 (d) 49

[SSC, 2014]

37. If x is a rational number and $\frac{(x+1)^3-(x-1)^3}{(x+1)^2-(x-1)^2}=2$, then the sum of numerator and denominator of x is

(a) 3 (b) 4
(c) 5 (d) 7

[SSC, 2014]

38. If $x=\sqrt{5}+2$, then the value $\frac{2x^2-3x-2}{3x^2-4x-3}$ is equal to

(a) 0.185 (b) 0.525
(c) 0.625 (d) 0.785

[SSC, 2014]

39. If $x^2 + y^2 + 1 = 2x$, then the value of $x^3 + y^5$ is

(a) 2 (b) 0
(c) –1 (d) 1

[SSC, 2014]

40. If $x^4 + \frac{1}{x^4} = 119$, then the value of $x^3-\frac{1}{x^3}$ is

(a) ±36 (b) ±33
(c) 36 (d) –36

[SSC, 2014]

41. If $\left\{\frac{1}{2}(a-b)\right\}^2+ab=p(a+b)^2$, then the value of p is (assume that $a \neq -b$)

(a) $\frac{1}{4}$ (b) $\frac{1}{8}$
(c) 1 (d) $\frac{1}{2}$

[SSC, 2014]

42. The reciprocal of $x+\frac{1}{x}$ is

(a) $x-\frac{1}{x}$ (b) $\frac{1}{x}+x$
(c) $\frac{x}{x^2+1}$ (d) $\frac{x}{x+1}$

[SSC, 2014]

43. If $x(x - 3) = -1$, then the value of $x^3(x^3 - 18)$ is

(a) 1 (b) 0
(c) –1 (d) 2

[SSC, 2014]

44. If $a\left(2+\sqrt{3}\right)=b\left(2-\sqrt{3}\right)=1$, then the value of $\frac{1}{a^2+1}+\frac{1}{b^2+1}$ is

(a) 4 (b) 9
(c) –5 (d) 1

[SSC, 2014]

45. Find the value of $\sqrt{30+\sqrt{30+\sqrt{30+\cdots}}}$.

(a) 5 (b) $3\sqrt{10}$
(c) 6 (d) 7

[SSC, 2013]

Directions (46–50): In each of these questions, two equations (I) and (II) are given. You have to solve both the equations and give answer

(a) If $x < y$ (b) If $x > y$
(c) If $x = y$ (d) If $x \geq y$
(e) If $x \leq y$ or no relationship can be established between x and y.

[IBPS PO/MT, 2013]

46. I. x^2–24x + 144 = 0
II. y^2– 26y + 169 = 0

47. I. 2x^2+ 3x– 20 = 0
II. 2y^2 + 19y + 44 = 0

48. I. 6x^2 + 77x + 121 = 0
II. y^2 + 9y – 22 = 0

49. I. x^2– 6x = 7
II. 2y^2 + 13y + 15 = 0

50. I. 10x^2– 7x + 1 = 0
II. 35y^2 –12y + 1 = 0

[IBPS PO/MT, 2013]

51. $\sqrt{6+\sqrt{6+\sqrt{6+\cdots}}}$ is equal to

(a) 2 (b) 5
(c) 4 (d) 3

[SSC, 2011]

52. The sum of the squares of two natural consecutive odd numbers is 394. The sum of the numbers is:

(a) 24 (b) 32
(c) 40 (d) 28

[SSC, 2011]

Directions (53–57): In the following questions, two equations numbered I and II are given. You have to solve both questions and give answer

(a) if $x > y$
(b) if $x \geq y$
(c) if $x < y$
(d) if $x \leq y$
(e) if $x = y$ or the relationship cannot be established.

53. I. $x^2 - 19x + 84 = 0$

II. $y^2 - 25y + 156 = 0$

[IOB PO, 2011]

54. I. $x^3 - 468 = 1729$

II. $y^2 - 1733 + 1564 = 0$

[IOB PO, 2011]

55. I. $\frac{9}{\sqrt{x}} + \frac{19}{\sqrt{x}} = \sqrt{x}$

II. $y^5 - \frac{(2\times14)^{11/12}}{\sqrt{y}} = 0$

[IOB PO, 2011]

56. I. $\sqrt{784}x + 1234 = 1486$

II. $\sqrt{1089}y + 2081 = 2345$

[IOB PO, 2011]

57. I. $\frac{12}{\sqrt{x}} - \frac{23}{\sqrt{x}} = 5\sqrt{x}$

II. $\frac{\sqrt{y}}{12} - \frac{5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

[IOB PO, 2011]

Directions (58–62): In each of these questions, two equations are given. You have to solve these equations and find out the values of x and y and

Give answer
(a) if $x < y$ (b) if $x > y$
(c) if $x \leq y$ (d) if $x \geq y$
(e) if $x = y$

[Andhra Bank PO, 2011]

58. I. $4x + 7y = 209$

II. $12x - 14y = -38$

[Andhra Bank PO, 2011]

59. I. $17x^2 + 48x = 9$

II. $13y^2 = 32y - 12$

[Andhra Bank PO, 2011]

60. I. $16x^2 + 20x + 6 = 0$

II. $10y^2 + 38y + 24 = 0$

[Andhra Bank PO, 2011]

61. I. $8x^2 + 6x = 5$

II. $12y^2 - 22y + 8 = 0$

[Andhra Bank PO, 2011]

62. I. $18x^2 + 18x + 4 = 0$

II. $12y^2 + 29y + 14 = 0$

[Andhra Bank PO, 2011]

Directions (63–29): In the following questions two equations numbered I and II are given. You have to solve both the equations and give answer

(a) $x > y$ (b) $x \geq y$
(c) $x < y$ (d) $x \leq y$
(e) $x = y$ or the relationship cannot be established

[Corporation Bank PO, 2011]

63. I. $x^2 - 11x + 24 = 0$

II. $2y^2 - 9y + 9 = 0$

[Corporation Bank PO, 2011]

64. I. $x^3 \times 13 = x^2 \times 247$

II. $y^{1/3} \times 14 = 294 \div y^{2/3}$

[Corporation Bank PO, 2011]

65. I. $\frac{12\times4}{x^{4/7}} - \frac{3\times4}{x^{4/7}} = x^{10/7}$

II. $y^3 + 783 = 999$

[Corporation Bank PO, 2011]

66. I. $\sqrt{500}x + \sqrt{402} = 0$

II. $\sqrt{360}y + (200)^{1/2} = 0$

[Corporation Bank PO, 2011]

67. I. $(17)^2 + 144 \div 18 = x$

II. $(26)^2 - 18\times21 = y$

[Corporation Bank PO, 2011]

Directions (68–72): In each of these questions, two equations are given. You have to solve these equations and find out the values of x and y and give answer

(a) $x < y$ (b) $x > y$
(c) $x \leq y$ (d) $x \geq y$
(e) $x = y$

68. **I.** $16x^2 + 20x + 6 = 0$

II. $10y^2 + 38y + 24 = 0$

[Punjab and Sind Bank PO, 2011]

69. **I.** $18x^2 + 18x + 4 = 0$

II. $12y^2 + 29y + 14 = 0$

[Punjab and Sind Bank PO, 2011]

70. **I.** $8x^2 + 6x = 5$

II. $12y^2 - 22y + 8 = 0$

[Punjab and Sind Bank PO, 2011]

71. **I.** $17x^2 + 48x = 9$

II. $13y^2 = 32y - 21$

[Punjab and Sind Bank PO, 2011]

72. **I.** $4x + 7y = 209$

II. $12x - 14y = -38$

[Punjab and Sind Bank PO, 2011]

Directions (73–77): In the following questions two equations numbered I and II are given. You have to solve both the equations and give answer

(a) if $x > y$ (b) if $x \geq y$

(c) if $x < y$ (d) if $x \leq y$

(e) if $x = y$ or the relationship cannot be established.

[Indian Bank PO, 2010]

73. **I.** $x^2 - 4 = 0$

II. $y^2 + 6y + 9 = 0$

[Indian Bank PO, 2010]

74. **I.** $x^2 - 7x + 12 = 0$

II. $y^2 + y - 12 = 0$

[Indian Bank PO, 2010]

75. **I.** $x^2 = 729$

II. $y = \sqrt{729}$

[Indian Bank PO, 2010]

76. **I.** $x^4 - 227 = 398$

II. $y^2 + 321 = 346$

[Indian Bank PO, 2010]

77. **I.** $2x^2 + 11x + 14 = 0$

II. $4y^2 + 12y + 9 = 0$

[Indian Bank PO, 2010]

ANSWER KEYS

EXERCISE–1

1. (c)	**2.** (a)	**3.** (a)	**4.** (c)	**5.** (a)	**6.** (b)	**7.** (b)	**8.** (c)	**9.** (a)	**10.** (c)	**11.** (d)	**12.** (b)	**13.** (a)
14. (c)	**15.** (b)	**16.** (b)	**17.** (b)	**18.** (d)	**19.** (a)	**20.** (c)	**21.** (d)	**22.** (b)	**23.** (d)	**24.** (a)	**25.** (b)	**26.** (b)
27. (d)	**28.** (c)	**29.** (c)	**30.** (c)	**31.** (a)	**32.** (a)	**33.** (a)	**34.** (d)	**35.** (a)	**36.** (d)	**37.** (d)	**38.** (a)	**39.** (c)
40. (a)	**41.** (a)	**42.** (b)	**43.** (c)	**44.** (c)	**45.** (d)	**46.** (c)	**47.** (a)	**48.** (a)				

EXERCISE–2

1. (b)	**2.** (d)	**3.** (d)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (d)	**11.** (b)	**12.** (c)	**13.** (b)
14. (c)	**15.** (c)	**16.** (d)	**17.** (b)	**18.** (b)	**19.** (b)	**20.** (a)	**21.** (b)	**22.** (c)	**23.** (d)	**24.** (b)	**25.** (d)	**26.** (d)
27. (b)	**28.** (a)	**29.** (c)	**30.** (e)	**31.** (c)	**32.** (d)	**33.** (d)	**34.** (a)	**35.** (c)	**36.** (c)	**37.** (b)	**38.** (c)	**39.** (d)
40. (c)	**41.** (a)	**42.** (c)	**43.** (a)	**44.** (d)	**45.** (c)	**46.** (a)	**47.** (d)	**48.** (e)	**49.** (b)	**50.** (d)	**51.** (d)	**52.** (d)
53. (d)	**54.** (b)	**55.** (e)	**56.** (a)	**57.** (a)	**58.** (e)	**59.** (a)	**60.** (b)	**61.** (c)	**62.** (d)	**63.** (b)	**64.** (c)	**65.** (d)
66. (c)	**67.** (c)	**68.** (b)	**69.** (d)	**70.** (c)	**71.** (a)	**72.** (e)	**73.** (a)	**74.** (b)	**75.** (d)	**76.** (e)	**77.** (c)	

EXPLANATORY ANSWERS

EXERCISE-I

1. (c) $Px^2 + 4x + 1 = 0$

Compare with $Ax^2 + Bx + C = 0$, we get

$$A = P,\ B = 4,\ C = 1$$

For real roots,

$B^2 - 4AC \geq 0 \Rightarrow 16 - 4P \geq 0$

$\Rightarrow 16 \geq 4P \Rightarrow P \leq 4.$

2. (a) The given equation is $2x^2 + Px + 4 = 0$

$\Rightarrow P(x) = 0$ where $P(x) = 2x^2 + Px + 4 = 0$

If 2 is a root of $P(x) = 0$, then $P(2) = 0$

$\Rightarrow 2(2)^2 + P(2) + 4 = 0 \Rightarrow 8 + 2P + 4 = 0$

$$\Rightarrow 2P = -12 \Rightarrow P = \frac{-12}{2} = -6$$

Hence the given equation is

$$2x^2 - 6x + 4 = 0 \Rightarrow 2x^2 - 2x - 4x + 4 = 0$$

$$\Rightarrow 2x(x-1) - 4(x-1) = 0$$

$$\Rightarrow 2(x-2)(x-1) = 0$$

$$\Rightarrow x = 1 \text{ or } x = 2$$

Hence second root is 1.

3. (a) The given equation is

$$x^2 - 5x + 6 = 0 \Rightarrow x^2 - 2x - 3x + 6 = 0$$
$$\Rightarrow x(x-2) - 3(x-2) = 0$$
$$\Rightarrow (x-2)(x-3) = 0$$
$$\Rightarrow x - 2 = 0 \text{ or } x - 3 = 0$$

Thus, the other root of the given quadratic equation is 2.

4. (c) $\sqrt{7}x^2 - 6x - 13\sqrt{7} = 0$

$$\Rightarrow \sqrt{7}x^2 - 13x + 7x - 13\sqrt{7} = 0$$

$$\Rightarrow x(\sqrt{7}x - 13) + \sqrt{7}(\sqrt{7}x - 13) = 0$$

$$\Rightarrow (x + \sqrt{7})(\sqrt{7}x - 13) = 0$$

$$\Rightarrow x + \sqrt{7} = 0 \text{ or } \sqrt{7}x - 13 = 0$$

$$\Rightarrow x = -\sqrt{7} \text{ or } x = \frac{13}{\sqrt{7}} = \frac{13\sqrt{7}}{7}$$

Thus, the two roots of given quadratic equation are

$-\sqrt{7}$ and $\frac{13\sqrt{7}}{7}$.

5. (a) The given quadratic equation is

$$3a^2x^2 - abx - 2b^2 = 0 \Rightarrow 3a^2x^2 - 3abx + 2abx - 2b^2 = 0$$
$$\Rightarrow 3ax(ax - b) + 2b(ax - b) = 0$$
$$\Rightarrow (ax - b)(3ax + 2b) = 0$$
$$\Rightarrow ax = b \text{ or } 3ax = -2b$$
$$\Rightarrow x = \frac{b}{a} \text{ or } x = -\frac{2b}{3a}.$$

6. (b) The given quadratic equation is

$$a^2x^2 - 3abx + 2b^2 = 0 \Rightarrow a^2x^2 - 2abx - abx + 2b^2 = 0$$
$$\Rightarrow ax(ax - 2b) - b(ax - 2b) = 0$$
$$\Rightarrow (ax - 2b)(ax - b) = 0$$
$$\Rightarrow ax - 2b = 0 \text{ or } ax - b = 0$$
$$\Rightarrow x = \frac{2b}{a} \text{ or } x = \frac{b}{a}$$

Thus, the two roots of the given quadratic equation are

$\frac{2b}{a}$ and $\frac{b}{a}$.

7. (b) Sum of the roots $= \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$

Product of the roots $= (\sqrt{2})(2\sqrt{2}) = 4$

Hence the required quadratic equation is x^2 = (sum of the roots) x + (Product of two roots) = 0

$$\Rightarrow x^2 - 3\sqrt{2x} + 4 = 0.$$

8. (c) The given quadratic equation is

$$ax^2 + (4a^2 - 3b)x - 12ab = 0$$

$$\Rightarrow ax^2 + 4a^2x - 3bx - 12ab = 0$$

$$\Rightarrow ax(x + 4a) - 3b(x + 4a) = 0$$

$$\Rightarrow (ax - 3b)(x + 4a) = 0$$

$$\Rightarrow ax - 3b = 0 \text{ or } x + 4a = 0$$

$$\Rightarrow x = \frac{3b}{a} \text{ or } x = -4a$$

Thus, the two roots of the given quadratic equation are

$-4a$ and $\frac{3b}{a}$.

9. (a) The required quadratic equation is x^2 – (sum of the roots) x + (product of the roots) = 0

$\Rightarrow x^2 - 6x - 16 = 0.$

10. (c) The given quadratic equation is

$3x^2 - 5x + P = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 3,\ b = -5,\ c = P$

If the given quadratic equation has equal roots then its discriminant = 0

$\Rightarrow b^2 - 4ac = 0 \Rightarrow (-5)^2 - 4(3)(P) = 0$

$\Rightarrow 25 - 12P = 0 \Rightarrow P = \frac{25}{12}$

11. (d) Since α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$,

$\therefore \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(-\frac{b}{a}\right)^2 - 2\frac{c}{a} = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}.$

12. (b) Since α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$

$\therefore \alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{a}{c}$

Now, $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$= \left(-\frac{b}{a}\right)^3 - 3\frac{c}{a}\left(\frac{-b}{a}\right)$

$= \frac{-b^3}{a} + \frac{3bc}{a^2} = \frac{-b^2 + 3abc}{a^3}$

$= \frac{b(3ac - b^2)}{a^3}$

13. (a) Since α and β are the roots of the equation $ax^2 + bx + c = 0$

$\therefore \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

Now, $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$

$= \frac{\left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right)}{\frac{c}{a}} = \frac{\frac{b^2}{a^2} - \frac{2c}{a}}{\frac{c}{a}}$

$= \frac{\frac{b^2 - 2ac}{a^2}}{\frac{c}{a}} = \frac{b^2 - 2ac}{ac}.$

14. (c) One root is $\sqrt{5}$

So the other root is $-\sqrt{5}$

$\therefore$ Sum of the roots is = 0

and product of the roots $= (\sqrt{5})(-\sqrt{5}) = -5$

$\therefore$ Required equation is x^2 – (sum of the roots)x + (product of the roots) = 0

$\Rightarrow x^2 - 5 = 0.$

15. (b) The equation $x^2 - px + q = 0$; $p, q \in R$ has no real root if

$B^2 < 4AC$

$\Rightarrow (-p)^2 < 4.1.q\ [\because A = 1, B = -p, C = q]$

$\Rightarrow p^2 < 4q.$

16. (b) The given quadratic equation is $x^2 + 5px + 16 = 0$(1)

Comparing it with $ax^2 + bx + c = 0$, we get $a = 1$, $b = 5p$, $c = 16$

If equation (1) has no real roots, then discriminant < 0

$\Rightarrow b^2 - 4ac < 0 \Rightarrow (5p)^2 - 4(1)(16) < 0$

$\Rightarrow 25p^2 - 64 < 0 \Rightarrow 25p^2 < 64$

$\Rightarrow p^2 < \frac{64}{25} \Rightarrow p^2 - \frac{64}{25} < 0$

$\Rightarrow \left(p - \frac{8}{5}\right)\left(p + \frac{8}{5}\right) < 0$

$\Rightarrow$ either $p - \frac{8}{5} > 0$ and $p + \frac{8}{5} < 0$

i.e., $p > \frac{8}{5}$ and $p < -\frac{8}{5}$,

which is not possible

or, $p - \frac{8}{5} < 0$ and $p + \frac{8}{5} > 0$

i.e., $p < \frac{8}{5}$ and $p > -\frac{8}{5}$ i.e., $-\frac{8}{5} < p < \frac{8}{5}$.

17. (b) We have $3z^2 + 5z + k$

Here $a = 3$, $b = 5$, $c = k$

$D = b^2 - 4ac = 25 - 12k$

For equal linear factors to exist, $D \geq 0$

i.e., $25 - 12k \geq 0 \Rightarrow 25 \geq 12k$

$\Rightarrow k \leq \frac{25}{12}$

Therefore, the set of real numbers $\leq \frac{25}{12}$ gives the set of value of k for which the given quadratic polynomial can be factored into the product of real linear factors.

18. (d) Putting $x = 3$, we get

$27 + 3(k - 1) + 9 = 0$

or $27 + 3k - 3 + 9 = 0$

or $3k = -33$ or $k = -11$.

19. (a) The given quadratic equation is $3x^2 - 10x + 3 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 3,\ b = -10,\ c = 3$

$\therefore$ Sum of the roots $= -\dfrac{b}{a} = \dfrac{10}{3}$

$\because$ One root $= \dfrac{1}{3}$

$\therefore$ The other root $= \dfrac{10}{3} - \dfrac{1}{3} = \dfrac{9}{3} = 3.$

20 . (c) $x^4 + 7x^2 + 16 = (x^4 + 8x^2 + 16) - x^2$

$= (x^2 + 4)^2 - x^2$

$= (x^2 + 4 + x)(x^2 + 4 - x)$

$= (x^2 + x + 4)(x^2 - x + 4).$

21. (d) $x^2 - 7x + 10 = 0 \Leftrightarrow (x - 5)(x - 2) = 0$

$\Leftrightarrow x = 5, 2$

$x^2 - 10x + 16 = 0 \Leftrightarrow (x - 8)(x - 2) = 0$

$\Leftrightarrow x = 8, 2$

$\therefore$ Common root is 2.

22. (b) Here $a = 1,\ b = p,\ c = q$

The roots of the equation $x^2 + px + q = 0$ are equal if

$b^2 - 4ac = 0 \Rightarrow p^2 - 4q = 0 \Rightarrow p^2 = 4q.$

23. (d) $x^2 - 6x + 5 = 0 \Leftrightarrow (x - 5)(x - 1) = 0$

$\Leftrightarrow x = 5$ or 1

Also $|x - 3| = 2 \Leftrightarrow x - 3 = 2$ or

$-(x - 3) = 2 \Leftrightarrow x = 5$ or $x = 1$

$\therefore x^2 - 6x + 5 = 0$ and $|x - 3| = 2$ are equivalent.

24. (a) Let the smaller part be x. Then the larger part = $16 - x$. Now

$2(16 - x)^2 - x^2 = 164 \Rightarrow 2(256 + x^2 - 32x) - x^2 = 164$

$\Rightarrow x^2 - 64x + 348 = 0$

$\Rightarrow x^2 - 6x - 58x + 348 = 0$

$\Rightarrow x(x - 6) - 58(x - 6) = 0$

$\Rightarrow (x - 6)(x - 58) = 0$

$x = 6$ or $x = 58$

But $m = 58$ is not possible, since sum of the two parts is 16

$\therefore$ $x = 6$, $\therefore$ other part = 10.

25. (b) The given equation is of the form

$ax^2 + bx + c = 0$

Also, $D = \sqrt{9} = 3$

So, roots are rational

Hence, both the roots must be integers.

26. (b) Given equation is

$2x - x^2 = x - 2 \Leftrightarrow x^2 - x - 2 = 0$

$\Leftrightarrow (x + 1)(x - 2) = 0$

$\Leftrightarrow x = 2$ or $= -1.$

27. (d) $\log_{10}(x^2 - 6x + 45) = 2$

$\Leftrightarrow x^2 - 6x + 45 = 10^2 = 100$

$\Leftrightarrow x^2 - 6x - 55 = 0$

$\Leftrightarrow (x - 11)(x + 5) = 0$

$\Leftrightarrow x = 11$ or $x = -5.$

28. (c) Comparing $x^2 - 5x + 6 = 0$ with

$ax^2 + bx + c = 0$

$a = 1, b = -5, c = 6$

$\therefore\ \alpha + \beta = \dfrac{-b}{a} = \dfrac{5}{1} = 5$

$\alpha\beta = \dfrac{c}{a} = 6$

Now, we are to form an equation whose roots are $\dfrac{1}{\alpha}, \dfrac{1}{\beta}$.

So the required equation is

$x^2 -$ (sum of roots)x + (Product of roots) = 0

$x^2 - \left(\dfrac{1}{\alpha} + \dfrac{1}{\beta}\right)x + \left(\dfrac{1}{\alpha} \cdot \dfrac{1}{\beta}\right) = 0$

$x^2 - \left(\dfrac{\alpha + \beta}{\alpha\beta}\right)x + \left(\dfrac{1}{\alpha\beta}\right) = 0$

$x^2 - \dfrac{5}{6}x + \dfrac{1}{6} = 0$

$6x^2 - 5x + 1 = 0.$

29. (c) Given equation is: $y + \dfrac{1}{y} = \dfrac{10}{3}$,

where $y = \dfrac{x + 4}{x - 4}$

$\therefore\ 3y^2 - 10y + 3 = 0 \Rightarrow y = 3, \dfrac{1}{3}$

$\therefore\ y = \dfrac{x+4}{x-4} = 3$ or, $y = \dfrac{x+4}{x-4} = \dfrac{1}{3}$

$3x + 12 = x - 4$ or, $x + 4 = 3x - 12$

$\Rightarrow\ x = -8$ or, $x = 8$

30. (c) For reciprocal roots, product of roots must be 1

$\therefore \quad \frac{c}{a} = 1$ i.e., $c = a$

31. (a) Sum of the roots = 6

One root $= 3 - \sqrt{5}$

$\therefore$ the other root $= 6 - (3 - \sqrt{5}) = 3 + \sqrt{5}$

$\therefore$ Product of roots $= (3 - \sqrt{5})(3 + \sqrt{5})$

$= 9 - 5 = 4$

Hence the required equation is

$x^2 -$ (sum of the roots)x + (product of roots) = 0

$\Rightarrow \quad x^2 - 6x + 4 = 0.$

32. (a) $\alpha + \beta = 6$ and $3\alpha + 2\beta = 20$

$\Rightarrow \quad \alpha = 4, \beta = 2$

Product of the roots = k

So, $k = \alpha\beta = 4 \times 2 = 8.$

33. (a) Let, the two consecutive odd positive integers be $2x + 1$ and $2x + 3$ where x is a whole number.

Now,

$(2x + 1)^2 + (2x + 3)^2 = 290$

$\Rightarrow \quad 4x^2 + 4x + 1 + 4x^2 + 12x + 9 = 290$

$\Rightarrow \quad 8x^2 + 16x - 280 = 0$

$\Rightarrow \quad x^2 + 2x - 35 = 0$

$\Rightarrow \quad (x + 7)(x - 5) = 0 \Rightarrow x = 7, -5$

But, $x = -7$ is not possible, since -7 is not a whole number

$\therefore \quad x = 5.$

34. (d) Let, the roots be α and $-\alpha$. Then, sum of roots = 0

Also, roots being not equal, discriminant $\neq 0$

$\therefore \frac{q}{p} = 0$ and $q^2 - 4pr \neq 0$

$\Leftrightarrow q = 0$ and $pr \neq 0.$

35. (a) Comparing $x^2 - 2(1 + 3x)x + 7(3 + 2k) = 0$ with $ax_2 + bx + c = 0$, we get

$a = 1, b = -2(1 + 3k), c = 7(3 + 2k)$

For equal roots $D = b^2 - 4ac = 0$

$\therefore 4(1 + 3k)^2 - 4 \times 1 \times 7(3 + 2k) = 0$

$\Rightarrow \quad 4(1 + k)^2 + 6k) - 84 - 56k = 0$

$\Rightarrow \quad 36k^2 - 32k - 80 = 0$

$\Rightarrow \quad 9k - 8k - 20 = 0$

$k = \frac{8 \pm \sqrt{64 - 4(9)(-20)}}{2 \times 9}$

$= \frac{8 \pm \sqrt{784}}{18} = \frac{8 \pm 28}{18}$

$= \frac{36}{18}, \frac{-20}{18} = 2, \frac{-10}{9}.$

36. (d) Let, α be a common root of the given equations.

Then, $\alpha^2 + 2\alpha - 3 = 0$ and $\alpha^2 + 3\alpha - k = 0$

$\therefore \quad \frac{\alpha^2}{-2k + 9} = \frac{\alpha}{-3 + k} = \frac{1}{3 - 2}$

So, $\alpha^2 = \frac{9 - 2k}{1}$ and $\alpha = \frac{k - 3}{1}$

So, $(9 - 2k) = (k - 3)^2$ or $k^2 - 4k = 0$

or, $k(k - 4) = 0$, so $k = 4.$

37. (d) Given equation is: $2^{2x} - 3 \cdot 2^x \times 2^2 + 32 = 0$

or, $2^{2x} - 12 \times 2^x + 32 = 0$

$\Rightarrow y^2 - 12y + 32 = 0$, where $2^x = y$

$\Rightarrow (y - 8)(y - 4) = 0 \Rightarrow y = 8, y = 4$

$\therefore \quad 2^x = 8$ or, $2^x = 4$

$\Rightarrow 2^x = 2^3$ or, $2^x = 2^2$

$\Rightarrow x = 3$ or, $x = 2.$

38. (a) Let, the roots be 3α and 2α

Then, $3\alpha + 2\alpha = \frac{m}{12} \Rightarrow \alpha = \frac{m}{60}$

$\therefore \quad \frac{5}{72} = \left(\frac{m}{60}\right)^2 \Leftrightarrow \frac{5}{72} = \frac{m^2}{3600}$

$\Leftrightarrow m^2 = \frac{3600 \times 5}{72} = 250$

$\therefore \quad m = \sqrt{250} = 5\sqrt{10}.$

39. (c) $\because$ α, β are the roots of the equation $2x^2 - 3x + 1 = 0$

$\therefore \quad \alpha + \beta = \frac{3}{2}$ …(1)

and $\alpha\beta = \frac{1}{2}$ …(2)

We are to form a quadratic equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ S = sum of the roots

$= \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$

$$= \frac{\left(\frac{3}{2}\right)^2 - 2\left(\frac{1}{2}\right)}{\frac{1}{2}} \qquad \text{[using (1) and (2)]}$$

$$= \frac{\frac{9}{4} - 1}{\frac{1}{2}} = \frac{5}{4} \times \frac{2}{1} = \frac{5}{2}$$

P = Product of the roots

$$= \frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 1$$

Hence the required quadratic equation is x^2 – (sum of the roots)x + (Product of the roots) = 0

$$\Rightarrow \quad x^2 - \frac{5}{2}x + 1 = 0$$

$$\Rightarrow \quad 2x^2 - 5x + 2 = 0.$$

40. (a) The given quadratic equation is

$3x^2 - 20x + 17 = 0$ (1)

Compare with $ax^2 + bx + c = 0$, we get

$a = 3,\ b = -20,\ c = 17$

The roots of (1) are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{20 \pm \sqrt{400 - 4(3)(17)}}{2 \times 3}$$

$$= \frac{20 \pm \sqrt{196}}{6} = \frac{20 + 14}{6}, \frac{20 - 14}{6}$$

$$= \frac{34}{6}, \frac{6}{6} = \frac{17}{3}, 1$$

Hence the roots of (1) are $\frac{17}{3}$ and 1. So we have to form an equation whose are $\frac{3}{17}$ and 1

Sum of the roots $= \frac{3}{17} + 1 = \frac{20}{17}$

Product of the roots $= \frac{3}{17} \times 1 = \frac{3}{17}$

Hence, the required equation is

$$x^2 - \frac{20}{17}x + \frac{3}{17} = 0 \Rightarrow 17x^2 - 20x + 3 = 0.$$

41. (a) $\because$ α, β are the roots of the equation $x^2 - 3\lambda x + \lambda^2 = 0$

$\alpha + \beta = 3\lambda$ …(1)

and $\alpha\beta = \lambda^2$ …(2)

Now, $\alpha^2 + \beta^2 = \frac{7}{4}$ (given)

$$\Rightarrow \quad (\alpha + \beta)^2 - 2\alpha\beta = \frac{7}{4}$$

$$\Rightarrow (3\lambda)^2 - 2\lambda^2 = \frac{7}{4} \Rightarrow 7\lambda^2 = \frac{7}{4}$$

$$\Rightarrow \lambda^2 = \frac{1}{4} \Rightarrow \lambda = \pm\frac{1}{2}.$$

42. (b) $\because$ α, β are the roots of the equation

$ax^2 + bx + b = 0$

$\therefore \alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{b}{a}$

Now, $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} = \frac{\alpha + \beta}{\sqrt{\alpha\beta}} + \sqrt{\frac{b}{a}}$

$$= \frac{-\frac{b}{a}}{\sqrt{\frac{b}{a}}} + \sqrt{\frac{b}{a}}$$

$$= -\sqrt{\frac{b}{a}} + \sqrt{\frac{b}{a}} = 0.$$

43. (c) Comparing $x^2 - x + 1$ with $ax^2 + bx + c$ we have

$a = 1,\ b = -1,\ c = 1$

Here $D = b^2 - 4ac = (-1)^2 - 4(1)\ (1)$

$= 1 - 4 = -3$

Since $D < 0$, so the given expression has no proper linear factor.

44. (c) Let, the breadth of the rectangular plot be x m. Then the length of rectangular plot = $(x + 8)$ m

$\therefore$ Area = Length × Breadth = $x(x + 8) m^2$ But the area of the plot is given to be 308 m^2

$$\therefore \quad x(x+8) = 308 \Rightarrow x^2 + 8x - 308 = 0$$

$$\Rightarrow x^2 + 22x - 14x - 308 = 0$$

$$\Rightarrow x(x+22) - 14(x-22) = 0$$

$$\Rightarrow (x+22)(x-14) = 0$$

$$\Rightarrow x = 14, -22$$

But, $x = -22$ is not possible, since breath cannot be negative

$\therefore \quad x = 14$

Hence the breadth of the rectangular plot = 14 m Length of the rectangular plot = (14 + 8) m = 22 m.

45. (d) Let, α, β be the roots of the equation $x^2 + kx + 12 = 0$

$\therefore \quad \alpha + \beta = -k$ and $\alpha\beta = 12$

Now $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$(1)^2 = k^2 - 4(12) \Rightarrow k^2 = 49 \Rightarrow k = \pm 7.$

46. (c) Put $x - \frac{1}{x} = y$

$$\therefore \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = \left(x + \frac{1}{x}\right)^2 + 4 = y^2 + 4$$

So, given equation becomes

$$\Rightarrow \quad y\left(y - \frac{3}{2}\right) = 0 \Rightarrow y = 0 \text{ or } y = \frac{3}{2}$$

$$\therefore \quad x - \frac{1}{x} = 0 \text{ or } x - \frac{1}{x} = \frac{3}{2}$$

$\Rightarrow x^2 - 1 = 0$ or $2x^2 - 3x - 2 = 0$

$\Rightarrow x = \pm 1 \quad$ or $(2x + 1)(x - 2) = 0$

or $x = -1/2$ or $x = 2$.

47. (a) $\because \quad \alpha, \beta$ are the roots of the equation $x^2 - 8x + 1 = 0$

$$\therefore \quad \alpha + \beta = \frac{-b}{a} = \frac{-(-8)}{1} = 8$$

and $\alpha\beta = \frac{c}{a} = \frac{k}{1} = k$

Now, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\Rightarrow 40 = (8)^2 - 2k \Rightarrow 2k = 24 \Rightarrow k = 12.$

48. (a) The given equation is $3x^2 + (2k + 1)x - k - 5 = 0$

Compare with $ax^2 + bx + c = 0$, we get

$a = 3, b = 2k + 1, c = -k - 5$

$$\therefore \quad \text{Sum of the roots} = \frac{-b}{a} = \frac{-(2k+1)}{3}$$

$$\text{and Product of the roots} = \frac{c}{a} = \frac{-k-5}{3} = \frac{-(k+5)}{3}$$

$\because$ Sum of the roots = Product of the roots

$$\therefore \frac{-(2k+1)}{3} = -\frac{(k+5)}{3} \Rightarrow 2k + 1 = k + 5$$

$\Rightarrow 2k - k = 5 - 1$

$\Rightarrow k = 4.$

EXERCISE–2
(BASED ON MEMORY)

1. (b) For equal roots

$a = b^2 - 4ac = 0$

Going by option

(a) $30 - 4(b) \neq 0$

(b) $36 - 4(a) = 0$

$3x^2 - 6x + 3 = 0$ has equal roots.

2. (d) Sum of roots $= \frac{-b}{a}$

$$1 = \frac{-b}{a}$$

Product of root $= \frac{c}{a}$

$$20 = \frac{c}{a}$$

Equation $= x^2 - x + 20 = 0$

3. (d) $a(x + y) = b(x - y) = 2ab$

$x + y = 2b$ (i)

$x - y = 2a$ (ii)

Adding (i) and (ii)

$x = a + b$

Subtracting (i) and (ii)

$y = b - a$

Now,

$2(x^2 + y^2)$

from (i) and (ii)

$2((a + b)^2 + (b - a)^2)$

$2(a^2 + b^2 + 2ab + a^2 + b^2 - 2ab)$

$2(2a^2 + 2b^2)$

$4(a^2 + b^2)$

4. (c) $(x - 2)(x - p) = x^2 - ax + 6$

$x^2 - px - 2x + 2p = x^2 - ax + 6$

$-px - 2x + 2p = ax + b$

Equating coefficients of x

$p + 2 = a$

$a - p = 2$

5. **(a)** $a^2 = by + cz$

$b^2 = cz + ax$

$c^2 = ax + by$

$a^2 + ax = ax + by + cz$

$a(a + x) = ax + by + cz$

$$\frac{1}{a+x} = \frac{a}{ax+by+cz}$$

Similarly

$$\frac{1}{b+x} = \frac{b}{ax+by+cz},$$

$$\frac{1}{c+x} = \frac{c}{ax+by+cz}.$$

$$\frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z}$$

$$\frac{ax}{ax+by+cz} + \frac{by}{ax+by+cz} + \frac{cz}{ax+by+cz}$$

$$\frac{ax+by+cz}{ax+by+cz} = 1$$

6. **(a)** Given $x = 332$, $y = 333$, $z = 335$

$$x^3 + y^3 + z^3 - 3xyz = (x+y+z)$$

$$(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$(x+y+z)\frac{1}{2}\left[(x-y)^2 + (y-z)^2 + (z-x)^2\right]$$

$$\frac{1}{2}[332+333+335][1+4+9]$$

$= 7000.$

7. **(d)** $m = -4$ and $n = -2$

$$m^3 - 3m^2 = 3m + 3n + 3n^2 + n^3$$

$$\Rightarrow \left(m^3 - 3m^3 + 3m - 1\right) + \left(n^3 + 3n^2 + 3n + 1\right)$$

$$\Rightarrow (m-1)^3 + (n+1)^3$$

$$= (-4-1)^3 + (-2+1)^3$$

$$\Rightarrow (-5)^3 + (-1)^3 \Rightarrow -125 - 1 \Rightarrow -126.$$

9. **(a)** $x = \dfrac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}}, y = \dfrac{1}{x} = \dfrac{\sqrt{13}+\sqrt{11}}{\sqrt{13}+\sqrt{11}}$

$$x + y = \frac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}} + \frac{\sqrt{13}+\sqrt{11}}{\sqrt{13}-\sqrt{11}}$$

$$= \left(\frac{B+11}{13-11}\right) \times 2 = \frac{24}{2} \times 2 = 24$$

$$x \cdot y = \frac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}} \times \frac{\sqrt{13}+\sqrt{11}}{\sqrt{13}-\sqrt{11}} = 1$$

$$3x^2 - 5xy + 3y^2 = 3(x+y)^2 - 11xy.$$

$$= 3(24)^2 - 11(1)$$

$$= 1717$$

10. **(d)** $x^2 + y^2 + z^2 - xy - yz - zx = 0$

$$\frac{1}{2}\left[(x-y)^2 + (y-z)^2 + (z-x)^2\right] = 0$$

$x = y = z = 1$

$$\therefore \frac{3x^4 + 7y^4 + 5z^4}{5x^2y^2 + 7y^2z^2 + 3z^2x^2} = \frac{3+7+5}{5+7+3} = 1.$$

11. **(b)** $x - \sqrt{3} - \sqrt{2} = 0$

$$\Rightarrow x = \sqrt{3} + \sqrt{2}$$

Again, $y = \sqrt{3} + \sqrt{2} = 0$

$$y = \sqrt{3} - \sqrt{2}$$

$$\therefore x - y = \sqrt{3} + \sqrt{2} - \sqrt{3} + \sqrt{2} = 2\sqrt{2}$$

and $xy = \left(\sqrt{3}+\sqrt{2}\right)\left(\sqrt{3}-\sqrt{2}\right) = 3 - 2 = 1$

$$\therefore \text{ Expression } \Rightarrow x^2 - 20\sqrt{2} - y^2 - 2\sqrt{2}$$

$$\Rightarrow x^3 - y^3 - 22\sqrt{2}$$

$$= (x-y)^3 + 3xy(x-y) - 22\sqrt{2}$$

$$= \left(2\sqrt{2}\right)^3 + 3\left(2\sqrt{2}\right) - 22\sqrt{2}$$

$$= 16\sqrt{2} + 6\sqrt{2} - 22\sqrt{2} = 0.$$

12. **(c)** $3\left(a^2 + b^2 + c^2\right) = (a+b+c)^2$

$$3a^2 + 3b^2 + 3c^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$$

$$\Rightarrow 3a^2 + 3b^2 + 3c^2 - a^2 - b^2 - c^2 = 2ab + 2bc + 2ac$$

$$\Rightarrow 2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ac = 0$$

$$\Rightarrow a^2 + b^2 - 2ab + b^2 + c^2 - 2bc + a^2 + c^2 - 2ac = 0$$

$$\Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 = 0$$

$\Rightarrow a - n = 0 \Rightarrow a = b$

$\Rightarrow b - c = 0 \Rightarrow b = c$

$\Rightarrow c - a = 0 \Rightarrow c = a$

$\therefore a = b = c.$

13. **(b)** $\dfrac{x^3 - y^3}{x^2 + xy + y^2} = \dfrac{5}{1}$

$$\frac{(x-y)\left(x^2 + xy + y^2\right)}{x^2 + xy + y^2} = 5$$

$$x - y = 5 \qquad (1)$$